

Geometry (once)

θ, α from the level set
absorb slivers, close their faces



Assemble momentum

θ -weighted ∂_t , conv., visc.
no wall term



Predictor

$$a_P \mathbf{u}^* = H(\mathbf{u}) - \nabla \nabla p^n$$



SIMPLEC split

$$r_{AU}, r_{AtU}, \mathbf{H}/a_P$$



Face fluxes

$$\phi_f = \theta_f \bar{\mathbf{u}}_f \cdot \mathbf{S}_f$$



Pressure

$$\nabla \cdot (r_{AU} \nabla \nabla p) = \nabla \cdot \phi$$

θ -weighted



Correct

$$\text{update } \phi, \text{ then } \mathbf{u} = \mathbf{H}/a_P - r_{AtU} \nabla \nabla p$$



Flow rate

two affine passes $\Rightarrow$ exact g_x

repeat to steady state